Mathematical Modeling Foundations — From Matrix Tricks to MBAM

Based on MIT 6.S087/18.03/18.031/18.335/18.337/6.441; MBAM/QSP

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Matrix, dynamics, geometry 2026

Reading motive

这篇笔记不再把 MIT 线性代数、ODE、LTI、数值计算、信息论/信息几何和 MBAM/QSP 当作相邻课程来摆放，而是把它们看成同一个对象的不同投影。这个对象就是

$$\theta \longmapsto \mathbf{y}(\theta),$$

the map from mechanisms to predictions. Everything below asks one question from different angles: how does a mechanistic parameter move the predictions we care about? [spine]

Audience

读者可以没有完整上过 6.S087、18.03、18.031、18.335 或 6.441，但需要愿意接受一个贯穿全篇的翻译："matrix" 是局部线性化，"ODE" 是产生预测的 state map，"conditioning" 是可解释性的数值体温计，"Fisher information" 是 model manifold 上的 metric。[from zero]

Style

This is a synthesis note rather than an excerpt. It keeps the dec41/pdf2htmlEX layout, uses the same notation discipline as the Population PK note, and tries to make every chapter hand its object to the next one. [workflow]

Contents

1 Notation and source map

1.1 Objects before formulas

The rule is simple: name the object before manipulating it. In this note,

$$\begin{aligned} \mathbf{x} \in \mathbb{R}^n &\quad \text{is a state or vector,} \\ \theta \in \mathbb{R}^p &\quad \text{is a parameter vector,} \end{aligned}$$

$$\begin{aligned} \mathbf{A} : \mathbb{R}^n \to \mathbb{R}^m &\quad \text{is a linear map,} \\ \mathbf{y}(\theta) \in \mathbb{R}^m &\quad \text{is a model prediction.} \end{aligned}$$

Random variables use sans-serif, for example x, while observed values use ordinary type, x. Parameters in classical fitting are ordinary unknowns, θ. If a fully Bayesian model treats them as random, I write $\boldsymbol{\theta}$.

Notation. Vectors and matrices are bold. Linear maps are not merely arrays of numbers; arrays are coordinate representations of maps. This distinction matters when we later ask which parameter combinations are identifiable independently of coordinates.

1.2 The source packet

The materials you supplied form a natural ladder:

Block	Source role	Pages
6.S087 Matrix Tricks	probability review, covariance, matrix identities, matrix calculus, Gaussian algebra	49
Matrix Cookbook	derivative identities and matrix reference grammar	72
18.03 Differential Equations	modeling by rates, linear/nonlinear ODEs, qualitative behavior	239
18.031 LTI Systems	exponentials, poles, transfer functions, convolution, step/delta response	97
18.335 Numerical Computing	finite precision, conditioning, direct/iterative methods, eigenvalue algorithms	729
18.337 Julia Computing	random matrices, fast multipole ideas, parallel prefix, computation as structure	160+
6.441 Information Theory	divergence, mutual information, local Fisher geometry, information projection	294
MBAM and CRISP papers	model manifolds, sloppiness, geodesic boundaries, QSP fit-for-purpose reduction	88

The exact page counts are less important than the conceptual order. The early notes teach how to compute the local map. The middle notes teach how dynamics creates the map. The later notes teach why the map becomes geometry. MBAM then asks what the geometry says we can remove.

2 The spine: one map, six lenses

2.1 One sentence

The whole path can be compressed into one sentence:

> A complex mechanistic model is a map from parameters to predictions; linear algebra studies the local derivative of that map, ODE/LTI theory gives its dynamical form, numerical analysis tells us when the derivative is trustworthy, information geometry turns the derivative into a metric, and MBAM follows that metric to simplify the model without losing the predictions we care about.

This sentence is dense, so the note expands it slowly.

2.2 The object that survives every translation

In the beginning, a model may look like chemistry, circuits, compartments, signaling pathways, or partial knowledge written as differential equations. After choosing a query, however, it always produces a prediction vector:

$$\begin{gathered} \theta \\ \downarrow \text{ mechanism} \\ \mathbf{x}(t;\theta) \\ \downarrow \text{ observation} \\ \mathbf{y}(\theta) \\ \downarrow \text{ comparison} \\ \mathbf{r}(\theta) \end{gathered}$$

This chain is the spine of the note. Linear algebra enters at $\partial \mathbf{y}/\partial\theta$. ODE enters because $\mathbf{x}(t;\theta)$ is generated by rates. LTI enters because local dynamics decomposes into modes and time scales. Numerical analysis enters because ill-conditioned derivatives can lie. Information geometry enters because a residual map gives parameter space a metric. MBAM enters because the metric has thin directions, and thin directions lead to boundaries.

2.3 A reader's map

The chapters therefore have a dependency, not merely an order:

$$\begin{gathered} \text{states and parameters} \\ \downarrow \\ \text{prediction map} \\ \downarrow \\ \text{Jacobian and singular directions} \\ \downarrow \\ \text{dynamical time scales} \\ \downarrow \\ \text{conditioning and reliable computation} \\ \downarrow \\ \text{Fisher metric and model manifold} \\ \downarrow \\ \text{boundary approximation} \end{gathered}$$

If a section ever feels abstract, return to this diagram and ask: which arrow is being clarified?

2.4 Why MBAM sits at the end

MBAM feels advanced because it presupposes many languages at once:

$$\begin{array}{c}\text{model}\\ \downarrow\\ \text{predictions}\\ \downarrow\\ \text{sensitivity matrix}\\ \downarrow\\ \text{metric}\\ \downarrow\\ \text{geodesic}\\ \downarrow\\ \text{limiting approximation}\end{array}$$

Each arrow is a course by itself. Matrix tricks explain the sensitivity matrix. ODE and LTI explain what a mechanistic model is. Numerical linear algebra explains why small singular values should make us cautious. Information theory explains why Fisher information is not arbitrary. MBAM then turns all of this into a model-reduction workflow.

3 Lens I: local linear structure

3.1 Coordinates are not the object

A vector $\mathbf{x} \in \mathbb{R}^n$ can be written as a column of numbers, but the deeper object is displacement, state, coefficient list, or perturbation. A matrix $\mathbf{A} \in \mathbb{R}^{m\times n}$ represents a linear map:

$$\mathbf{x} \mapsto \mathbf{A}\mathbf{x}.$$

The key property is superposition:

$$\mathbf{A}(a\mathbf{x} + b\mathbf{z}) = a\mathbf{A}\mathbf{x} + b\mathbf{A}\mathbf{z}.$$

This is the first reason linear algebra keeps returning in nonlinear modeling: every smooth nonlinear model becomes linear after zooming in far enough. The matrix is not the model; it is the model's local handwriting.

3.2 Four spaces

For a matrix $\mathbf{A}$, the four fundamental spaces are:

$$\mathrm{range}(\mathbf{A}), \quad \mathrm{null}(\mathbf{A}), \quad \mathrm{range}(\mathbf{A}^\top), \quad \mathrm{null}(\mathbf{A}^\top).$$

In least squares, the residual $\mathbf{r} = \mathbf{b} - \mathbf{A}\widehat{\mathbf{x}}$ is orthogonal to the column space:

$$\mathbf{A}^\top\mathbf{r} = 0.$$

This innocent equation is already a prototype for parameter estimation. Later, $\mathbf{A}$ becomes the Jacobian $J = \partial\mathbf{y}/\partial\theta$, and the normal equation becomes

$$J^\top\mathbf{r} = 0.$$

Remark. Parameter fitting is geometry before it is statistics: it asks for the point on a prediction surface closest to the data.

4 Lens II: what counts as a change

4.1 Norms encode what matters

A norm measures size:

$$\|\mathbf{x}\|_2 = \sqrt{\mathbf{x}^\mathsf{T}\mathbf{x}}, \qquad \|\mathbf{x}\|^2_{\mathbf{W}} = \mathbf{x}^\mathsf{T}\mathbf{W}\mathbf{x}.$$

The weighted norm is crucial. If two experimental measurements have different uncertainty, the model should not treat their errors equally. A residual vector $\mathbf{r}$ with covariance Σ naturally uses

$$\|\mathbf{r}\|^2_{\Sigma^{-1}} = \mathbf{r}^\mathsf{T}\Sigma^{-1}\mathbf{r}.$$

This is the least-squares version of likelihood geometry. A norm is therefore a scientific choice: it says which prediction differences matter and which differences are below the measurement scale.

4.2 The singular value decomposition

The SVD writes

$$\mathbf{A} = \mathbf{U}\Sigma\mathbf{V}^\mathsf{T}.$$

The right singular vectors $\mathbf{v}_j$ are input directions. The left singular vectors $\mathbf{u}_j$ are output directions. The singular values σ_j say how much $\mathbf{A}$ stretches:

$$\mathbf{A}\mathbf{v}_j = \sigma_j\mathbf{u}_j.$$

When σ_j is tiny, a large input motion produces almost no output motion. In parameter estimation, that means a parameter combination is weakly visible to the data.

4.3 Projection as compression

If $\mathbf{U}_k$ contains the first k left singular vectors, then

$$\mathbf{P}_k = \mathbf{U}_k\mathbf{U}_k^\mathsf{T}$$

projects onto a low-dimensional output space. This is the algebraic ancestor of model reduction: if most output variation lies in a few directions, we should ask whether the full model is carrying more complexity than the prediction task requires. The same question will reappear later as a model-manifold question.

5 The hinge: from nonlinear model to Jacobian

5.1 Jacobians

Let a model make predictions

$$\mathbf{y}(\theta) = (y_1(\theta), \ldots, y_m(\theta))^\mathsf{T}.$$

The Jacobian is

$$J(\theta) = \frac{\partial \mathbf{y}}{\partial \theta} = \begin{bmatrix} \frac{\partial y_1}{\partial \theta_1} & \cdots & \frac{\partial y_1}{\partial \theta_p} \\ \vdots & & \vdots \\ \frac{\partial y_m}{\partial \theta_1} & \cdots & \frac{\partial y_m}{\partial \theta_p} \end{bmatrix}.$$

Locally,

$$\mathbf{y}(\theta + \delta\theta) \approx \mathbf{y}(\theta) + J(\theta)\delta\theta.$$

This is the central bridge from nonlinear science to linear algebra. Every later object is a refinement of this line: the SVD asks which $\delta\theta$ matter, the Fisher metric weights those changes by uncertainty, and MBAM follows the least visible directions beyond the local neighborhood.

5.2 Gradient, Hessian, Gauss–Newton

For least squares,

$$\mathcal{L}(\theta) = \frac{1}{2}\left\|\mathbf{r}(\theta)\right\|^2, \qquad \mathbf{r}(\theta) = \mathbf{y}(\theta) - \mathbf{d}.$$

The gradient is

$$\nabla\mathcal{L}(\theta) = J(\theta)^\mathsf{T}\mathbf{r}(\theta).$$

The Hessian has two kinds of terms:

$$\nabla^2\mathcal{L} = J^\mathsf{T}J + \sum_{m=1}^{M} r_m \nabla^2 y_m.$$

Near a good fit, residuals are small, so the Gauss–Newton approximation uses

$$\nabla^2\mathcal{L} \approx J^\mathsf{T}J.$$

This approximation is also the deterministic least-squares form of the Fisher information matrix. Thus the Hessian is not only an optimizer's tool; in the right scaling it is the local geometry of distinguishability.

6 Lens III: dynamics creates the prediction map

6.1 Modeling by rates

An ODE writes instantaneous change as a function of current conditions:

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbf{x}(t) = \mathbf{f}(\mathbf{x}(t), u(t), \theta), \qquad \mathbf{x}(0) = \mathbf{x}_0(\theta).$$

In pharmacology and systems biology, $\mathbf{x}$ may contain concentrations, receptor occupancies, signaling states, or disease states. The model prediction is rarely the full state. Usually,

$$\mathbf{y}(\theta) = \mathbf{H}\mathbf{x}(t_{1:M}; \theta)$$

or some nonlinear observation map of sampled states.

6.2 The flow map

The ODE defines a flow:

$$\Phi_t(\mathbf{x}_0, \theta) = \mathbf{x}(t; \mathbf{x}_0, \theta).$$

Then a dynamic model is still a map from parameters to predictions:

$$\theta \longmapsto \mathbf{y}(\theta).$$

The fact that this map is produced by solving differential equations changes the computation, but not the geometry. We can still ask for $J = \partial\mathbf{y}/\partial\theta$, singular values, residuals, and parameter combinations.

6.3 Sensitivity equations

For each parameter θ_k, define a sensitivity vector

$$\mathbf{s}_k(t) = \frac{\partial \mathbf{x}(t)}{\partial \theta_k}.$$

Differentiating the ODE gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbf{s}_k = \frac{\partial \mathbf{f}}{\partial \mathbf{x}}\mathbf{s}_k + \frac{\partial \mathbf{f}}{\partial \theta_k}.$$

So the Jacobian of a dynamical model is itself generated by an ODE. This is why ODE, matrix calculus, and numerical analysis cannot be separated in MBAM. The object MBAM studies is geometric, but the geometry is generated by integrating dynamical sensitivities.

7 Lens IV: modes, poles, and time scales

7.1 Exponentials and eigenmodes

An LTI state-space model has the form

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}, \qquad \mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u}.$$

With no input,

$$\mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{x}(0).$$

If $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$, then the mode $\mathbf{v}$ evolves like $e^{\lambda t}$. Poles and eigenvalues are therefore time-scale statements:

$$\operatorname{Re}\lambda < 0 \quad \text{means decay,} \qquad |\operatorname{Re}\lambda|^{-1} \quad \text{is a characteristic time scale.}$$

7.2 Convolution and impulse response

For zero initial condition, the output of an LTI system is a convolution:

$$y(t) = (h * u)(t) = \int_0^t h(t-\tau)u(\tau)\,\mathrm{d}\tau.$$

The impulse response $h(t)$ is the system's fingerprint in time. In the Laplace domain,

$$Y(s) = H(s)U(s).$$

The transfer function $H(s)$ turns differential equations into algebraic structure. This is why 18.031 is so important for MBAM: simplification often means discovering that some poles, zeros, or pathways have little effect on the predictions of interest. A small singular direction often has a time-scale story hiding behind it.

7.3 A pole is a behavioral scale

Suppose

$$H(s) = \frac{b}{(s+\alpha)(s+\beta)}.$$

If $\beta \gg \alpha$, the mode $e^{-\beta t}$ decays very fast. For observations taken after that fast transient,

$$H(s) \approx \frac{b/\beta}{s+\alpha}.$$

This is already a boundary approximation: one time scale has gone to infinity, leaving an effective parameter b/β. Later, MBAM will find such limits by geometry rather than by guessing them from the transfer function.

8 Lens V: trustworthy computation

8.1 Conditioning

The condition number of a nonsingular matrix is

$$\kappa_2(\mathbf{A}) = \|\mathbf{A}\|_2 \left\|\mathbf{A}^{-1}\right\|_2 = \frac{\sigma_{\max}}{\sigma_{\min}}.$$

If κ is large, small data perturbations can create large solution perturbations. In parameter estimation, a large condition number of $J^\top J$ means the model has directions that the data barely sees.

Warning. Ill-conditioning is not only a numerical nuisance. In mechanistic modeling it is often a scientific signal: the model's predictions do not depend strongly on some parameter combinations.

8.2 Residual versus error

For a linear system $\mathbf{Ax} = \mathbf{b}$, the residual is

$$\mathbf{r} = \mathbf{b} - \mathbf{A}\widehat{\mathbf{x}}.$$

The error is

$$\mathbf{e} = \mathbf{x} - \widehat{\mathbf{x}}.$$

A small residual does not guarantee a small error when the problem is ill-conditioned. This distinction reappears in nonlinear model fitting: a model may fit observed data well while leaving many parameter directions unresolved.

8.3 QR, SVD, and least squares

QR solves least squares stably by orthogonalizing columns:

$$\mathbf{A} = \mathbf{QR}.$$

SVD is more diagnostic:

$$\widehat{\mathbf{x}} = \sum_{\sigma_j > 0} \frac{\mathbf{u}_j^\top \mathbf{b}}{\sigma_j} \mathbf{v}_j.$$

Tiny σ_j amplify noise. Truncated SVD removes directions that cannot be supported by the data. This is a linear version of fit-for-purpose reduction. MBAM is the nonlinear, mechanistic version: do not merely truncate the direction; ask what limiting model it points toward.

9 Large models and useful subspaces

9.1 Krylov thinking

Large systems are often solved without forming all matrices explicitly. Krylov methods build subspaces

$$\mathcal{K}_k(\mathbf{A}, \mathbf{r}_0) = \mathrm{span}\{\mathbf{r}_0, \mathbf{A}\mathbf{r}_0, \ldots, \mathbf{A}^{k-1}\mathbf{r}_0\}.$$

Conjugate gradient solves symmetric positive definite systems by choosing search directions that are conjugate under the $\mathbf{A}$-inner product.

The moral for modeling is simple: computation often exploits the fact that useful information lives in a much smaller subspace than the formal dimension suggests.

9.2 Preconditioning

A preconditioner $\mathbf{M}$ changes the geometry of the linear system:

$$\mathbf{M}^{-1}\mathbf{A}\mathbf{x} = \mathbf{M}^{-1}\mathbf{b}.$$

Good preconditioning clusters eigenvalues and reduces effective anisotropy. In parameter inference, reparameterization and nondimensionalization play an analogous role. Bad coordinates can make a model look harder than it is; good coordinates expose the meaningful combinations.

9.3 Parallel prefix as a small metaphor

The parallel prefix operation computes cumulative structure using associativity. The reason it matters here is not the particular algorithm, but the mental habit:

global computation   can be reorganized through algebraic structure.

QSP models are often large not because every piece is equally important, but because the raw representation has not yet exposed the correct structure. This is the computational version of the same refrain: the ambient dimension is usually larger than the active dimension.

10 Lens VI: distinguishability as geometry

10.1 Divergence

Information theory begins with comparing probability laws. The Kullback–Leibler divergence is

$$\mathrm{D}(P\|Q) = \mathbb{E}_P\left[\log \frac{p(\mathsf{x})}{q(\mathsf{x})}\right].$$

It is not symmetric and not a distance, but it measures distinguishability. In modeling, two parameter values θ and $\theta + \delta\theta$ are close if the probability laws they induce are hard to distinguish.

10.2 Local behavior and Fisher information

For a parametric family $p(x \mid \theta)$, the local expansion is

$$\mathrm{D}(p(\cdot \mid \theta) \,\|\, p(\cdot \mid \theta + \delta\theta)) = \frac{1}{2}\delta\theta^{\mathsf{T}}\mathcal{I}(\theta)\delta\theta + o(\|\delta\theta\|^2),$$

where

$$\mathcal{I}_{ij}(\theta) = \mathbb{E}_\theta\left[\frac{\partial \log p(\mathsf{x} \mid \theta)}{\partial\theta_i}\frac{\partial \log p(\mathsf{x} \mid \theta)}{\partial\theta_j}\right].$$

Thus Fisher information is a metric tensor: it says how parameter displacements translate into distinguishable changes in predictions. This is where the algebraic Jacobian becomes a geometric object.

10.3 Least squares as Fisher geometry

If observations are Gaussian with independent standard deviations σ_m,

$$d_m = y_m(\theta) + \sigma_m\varepsilon_m, \qquad \varepsilon_m \sim \mathcal{N}(0, 1),$$

then

$$\mathcal{I}(\theta) = J(\theta)^{\mathsf{T}}J(\theta), \qquad J_{mi} = \frac{1}{\sigma_m}\frac{\partial y_m}{\partial\theta_i}.$$

This is the exact bridge from numerical least squares to information geometry. The same matrix $J^{\mathsf{T}}J$ has now worn three costumes: curvature for optimization, conditioning for numerical analysis, and metric for statistical distinguishability.

11 The inverse problem: sloppiness has meaning

11.1 The residual map

Define the scaled residual map

$$\mathbf{r}(\theta) = \begin{bmatrix} (y_1(\theta) - d_1)/\sigma_1 \\ \vdots \\ (y_M(\theta) - d_M)/\sigma_M \end{bmatrix}.$$

Fitting minimizes

$$\frac{1}{2} \|\mathbf{r}(\theta)\|^2 .$$

The Jacobian of $\mathbf{r}$ determines the local geometry of the fitting problem. Its singular values tell us which parameter combinations are visible.

11.2 Sloppy spectra

A sloppy model has Fisher eigenvalues spread across many orders of magnitude:

$$\lambda_1 \gg \lambda_2 \gg \cdots \gg \lambda_p.$$

The eigenvectors with large eigenvalues are stiff directions: predictions change quickly. Eigenvectors with tiny eigenvalues are sloppy directions: parameters can move a lot while predictions barely change.

Remark. Sloppiness does not mean the model is useless. Often it means the model makes robust predictions even though many microscopic parameters remain uncertain.

11.3 Identifiable combinations

An identifiable parameter may not be a single named rate constant. It may be a combination:

$$\phi = \frac{k_{\text{cat}} E_0}{K_M}, \qquad \tau = \frac{1}{k_{\text{slow}}}, \qquad \rho = \frac{k_1}{k_2}.$$

This is why matrix derivatives and model structure must be read together. Linear algebra finds a direction; scientific interpretation turns it into a mechanism. Without the mechanism, sloppiness is only a spectrum. With the mechanism, it becomes a clue about what the data actually controls.

12 The model manifold: the same map seen globally

12.1 Prediction space

Fix a set of quantities of interest:

$$\mathcal{Q} = \{y_1, \dots, y_M\}.$$

The model maps parameter values into prediction space:

$$\theta \in \mathbb{R}^p \longmapsto \mathbf{y}(\theta) \in \mathbb{R}^M.$$

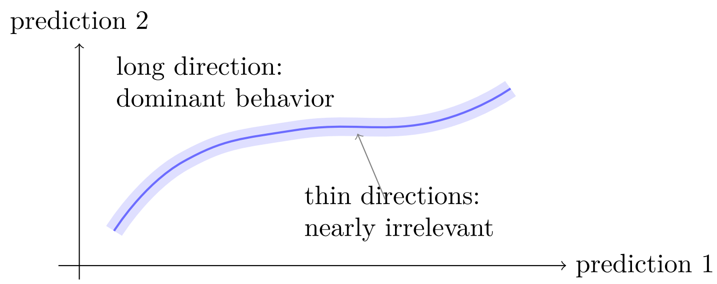
The image of this map is the model manifold:

$$\mathcal{M} = \{\mathbf{y}(\theta) : \theta \in \Theta\}.$$

The model manifold is not the space of mechanisms. It is the space of behaviors the mechanism can produce under the chosen query. This sentence is crucial: changing the query changes the manifold, even if the equations stay fixed.

12.2 Hyper-ribbon intuition

Many mechanistic models produce model manifolds shaped like hyper-ribbons: long in a few directions, thin in many others. The long directions correspond to behavior that can vary substantially. The thin directions correspond to parameter changes that create tiny changes in the quantities of interest.



12.3 Boundaries as approximations

If a manifold is thin, its boundary may be a good lower-dimensional approximation. Algebraically, boundaries often correspond to limits:

$$k \to 0, \qquad k \to \infty, \qquad \frac{k_1}{k_2} \to K, \qquad \tau_{\text{fast}} \to 0.$$

Scientifically, these limits are familiar: irreversible reactions, equilibrium approximations, quasi-steady state approximations, continuum limits, thermodynamic limits, and fast-mode elimination. So a boundary is not a mathematical accident. It is often an old modeling approximation rediscovered by asking what the data cannot distinguish.

13 MBAM: walking from weak direction to simpler mechanism

13.1 The algorithm

The Manifold Boundary Approximation Method can be summarized as:

(i) Choose the quantities of interest $\mathcal{Q}$.

(ii) Compute the Fisher information $\mathcal{I}(\theta)$.

(iii) Find the least informative eigenvector.

(iv) Follow a geodesic in that direction until the model manifold reaches a boundary.

(v) Identify the limiting approximation in the original equations.

(vi) Replace the full model by the boundary model and refit.

(vii) Repeat until the model is simple enough for the scientific question.

The important point is that MBAM removes parameter combinations, not merely named parameters. It begins with the same local matrix story as SVD, but it refuses to stop at a local vector; it walks until the equations reveal a limit.

13.2 Geodesic equation

On a manifold with metric g_{ij}, a geodesic satisfies

$$\ddot{\theta}^i + \Gamma^i_{jk}\dot{\theta}^j\dot{\theta}^k = 0.$$

Here g_{ij} is the Fisher metric. The geodesic starts in the least sensitive direction. As it approaches a boundary, the limiting form often becomes interpretable: one rate goes to zero, two rates go to infinity with fixed ratio, a fast variable equilibrates, or two modes merge.

Remark. The geodesic is a numerical flashlight. It does not itself explain the biology; it shows where to look in the equations for the simplifying limit.

13.3 Why this is not black-box reduction

Many reduction methods produce a smaller system that predicts well but is hard to interpret. MBAM tries to preserve mechanistic meaning by taking limits inside the original model. The reduced model remains expressed in terms of original mechanisms or combinations of them.

That is why MBAM is attractive for QSP: a regulatory or pharmacological model must not only predict. It must explain what biological simplification has been made.

14 Boundary examples: the abstractions become mechanisms

14.1 Sum of exponentials

Consider

$$y(t;\theta) = \sum_{j=1}^{N} A_j e^{-\lambda_j t}.$$

If one rate is much faster than all observed sampling times,

$$\lambda_j \to \infty, \qquad A_j e^{-\lambda_j t} \approx 0 \quad (t > 0).$$

If one rate is much slower than the experiment,

$$\lambda_j \to 0, \qquad A_j e^{-\lambda_j t} \approx A_j.$$

If two rates merge,

$$A_1 e^{-\lambda t} + A_2 e^{-\lambda t} = (A_1 + A_2) e^{-\lambda t}.$$

These are manifold boundaries with simple physical meaning: unobserved fast decay, constant slow background, and merged modes. Notice the full chain: LTI gives exponentials, SVD notices indistinguishable combinations, the manifold picture calls them boundaries, and MBAM turns the boundary into a reduced model.

14.2 Michaelis–Menten

The enzyme scheme

$$E + S \underset{k_r}{\overset{k_f}{\rightleftharpoons}} C \xrightarrow{k_c} E + P$$

has a boundary where $k_f, k_r \to \infty$ with $K_d = k_r/k_f$ fixed. The fast binding step equilibrates:

$$C \approx \frac{ES}{K_d}.$$

Using conservation of enzyme, the product rate becomes

$$\frac{\mathrm{d}}{\mathrm{d}t} P = \frac{k_c E_0 S}{K_d + S}.$$

MBAM recovers this familiar approximation geometrically: the data cannot distinguish the two fast rates separately, only their effective combination.

14.3 An LTI reduction

Suppose an LTI model contains a fast pole $-\beta$ and a slow pole $-\alpha$, with $\beta \gg \alpha$. If the query ignores very early transients, then the fast mode is a thin direction in prediction space. A boundary approximation removes it:

$$e^{-\beta t} \to 0 \quad \text{on the query window.}$$

The reduced model is not universally equivalent to the full model. It is equivalent relative to the quantities of interest. This phrase, relative to the quantities of interest, is the heartbeat of MBAM.

15 QSP and CRISP: the query closes the loop

15.1 QSP as query-driven modeling

QSP models become large because they collect plausible mechanisms. But a decision rarely needs every mechanism at full resolution. A toxicology query, a dose-selection query, and a target-discovery query may require different reduced models.

Let training data be $\mathcal{D}$ and future quantities of interest be $\mathcal{Q}$. The modeling problem is not merely

$$\text{fit } \mathcal{D}.$$

It is

$$\text{extract from } \mathcal{D} \text{ the information needed for } \mathcal{Q}.$$

This is the conceptual bridge from information geometry to QSP workflow. A model is not reduced in the abstract; it is reduced for a decision.

15.2 CRISP language

The CRISP idea can be read as:

$$\begin{array}{c} \text{comprehensive model} \\ \downarrow \\ \text{calibrated model} \\ \downarrow \\ \text{query-specific active manifold} \\ \downarrow \\ \text{fit-for-purpose reduced model} \end{array}$$

The reduction is contextual. It is not saying the removed mechanism is false. It is saying the mechanism is irrelevant to the specified prediction at the specified precision.

15.3 Mechanistic control knobs

The best reduced model does more than save computation. It identifies control knobs:

$$\phi_1(\theta), \ldots, \phi_k(\theta)$$

such that the quantities of interest depend mainly on ϕ, not on all microscopic parameters. In systems biology, these combinations may correspond to time-scale ratios, saturation constants, feedback strengths, or pathway capacities. This is the end of the chain that began with a Jacobian: a singular direction becomes a mechanistic control knob.

16 Detailed worked notebook: the same spine in slow motion

The following workbook is not a second, unrelated set of chapters. It repeats the same spine at lower altitude. The first group builds the linear and statistical geometry. The second group shows how ODE/LTI systems create the relevant sensitivities and time scales. The third group turns those sensitivities into Fisher geometry and model manifolds. The final group shows how boundaries become MBAM and CRISP reductions. Read it as a spiral: each example returns to

$$\begin{array}{c} \theta \longmapsto \mathbf{y}(\theta) \longmapsto J(\theta) \\ \downarrow \\ \mathcal{I}(\theta) \longmapsto \text{boundary} \end{array}$$

The worked notebook repeats the main spine at slower speed. The first pass is purely linear: covariance gives shape, projection gives fitting, derivatives give the local map, and the SVD tells us which combinations are visible.

16.1 Covariance as geometry

Why covariance is a matrix

For a random vector $\mathsf{x} \in \mathbb{R}^n$, covariance is

$$\mathrm{Cov}(\mathsf{x}) = \mathbb{E}\left[(\mathsf{x}-\mu)(\mathsf{x}-\mu)^\mathsf{T}\right].$$

The diagonal entries are variances. The off-diagonal entries are co-movements. But the geometric meaning is stronger: covariance defines the shape of uncertainty. If

$$\mathsf{x} \sim \mathcal{N}(\mu, \Sigma),$$

then contours of equal density are ellipsoids:

$$(\mathbf{x}-\mu)^\mathsf{T}\Sigma^{-1}(\mathbf{x}-\mu) = c.$$

Large eigenvalues of Σ are broad uncertainty directions. Small eigenvalues are tightly known directions.

The duality with information

In a regular Gaussian approximation, the parameter covariance is approximately the inverse Fisher information:

$$\mathrm{Cov}(\widehat{\theta}) \approx \mathcal{I}(\widehat{\theta})^{-1}.$$

So the eigenvectors of Σ and $\mathcal{I}$ point in the same directions, but their eigenvalues invert. A large covariance direction is a small information direction. This is one of the cleanest bridges between statistics and numerical linear algebra.

16.2 Projection and least squares

The nearest point problem

Given $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{b} \in \mathbb{R}^{m}$, least squares asks for

$$\widehat{\mathbf{x}} = \operatorname{argmin}_{\mathbf{x}} \|\mathbf{A}\mathbf{x} - \mathbf{b}\|_2^2 .$$

If $\mathbf{A}$ has full column rank, the solution satisfies

$$\mathbf{A}^{\mathsf{T}}\mathbf{A}\widehat{\mathbf{x}} = \mathbf{A}^{\mathsf{T}}\mathbf{b}.$$

This is not magic. At the closest point, the residual must be orthogonal to every column direction:

$$\mathbf{A}^{\mathsf{T}}(\mathbf{A}\widehat{\mathbf{x}} - \mathbf{b}) = 0.$$

Weighted projection

If measurements have covariance Σ, the geometry changes:

$$\widehat{\mathbf{x}} = \operatorname{argmin}_{\mathbf{x}} (\mathbf{A}\mathbf{x} - \mathbf{b})^{\mathsf{T}} \Sigma^{-1} (\mathbf{A}\mathbf{x} - \mathbf{b}).$$

The normal equations become

$$\mathbf{A}^{\mathsf{T}}\Sigma^{-1}\mathbf{A}\widehat{\mathbf{x}} = \mathbf{A}^{\mathsf{T}}\Sigma^{-1}\mathbf{b}.$$

Whitening with $\Sigma^{-1/2}$ turns the weighted problem into an ordinary Euclidean projection. This is exactly what scaled residuals do in model fitting.

16.3 Matrix derivatives

Differentials first

Matrix calculus is least confusing when written with differentials. If

$$f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x},$$

then

$$\mathrm{d}f = \frac{1}{2}\,\mathrm{d}\mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x} + \frac{1}{2}\mathbf{x}^{\mathsf{T}}\mathbf{A}\,\mathrm{d}\mathbf{x}.$$

If $\mathbf{A}$ is symmetric,

$$\mathrm{d}f = (\mathbf{A}\mathbf{x})^{\mathsf{T}}\,\mathrm{d}\mathbf{x}, \qquad \nabla f = \mathbf{A}\mathbf{x}.$$

The Hessian is $\mathbf{A}$.

Trace trick

The trace identity

$$\operatorname{tr}(\mathbf{A}\mathbf{B}) = \operatorname{tr}(\mathbf{B}\mathbf{A})$$

lets us move differentials to the end. For

$$f(\mathbf{X}) = \frac{1}{2}\|\mathbf{A}\mathbf{X} - \mathbf{B}\|_F^2 ,$$

we have

$$\mathrm{d}f = \operatorname{tr}\left[(\mathbf{A}\mathbf{X} - \mathbf{B})^{\mathsf{T}}\mathbf{A}\ \mathrm{d}\mathbf{X}\right],$$

so

$$\nabla_{\mathbf{X}} f = \mathbf{A}^{\mathsf{T}}(\mathbf{A}\mathbf{X} - \mathbf{B}).$$

This is the same normal-equation geometry in matrix form.

16.4 Whitening and likelihood

Gaussian residuals

Suppose

$$\mathbf{d} = \mathbf{y}(\theta) + \epsilon, \qquad \epsilon \sim \mathcal{N}(0, \Sigma).$$

The negative log likelihood, up to constants, is

$$\mathcal{L}(\theta) = \frac{1}{2}(\mathbf{d} - \mathbf{y}(\theta))^{\mathsf{T}} \Sigma^{-1} (\mathbf{d} - \mathbf{y}(\theta)).$$

Let

$$\widetilde{\mathbf{r}}(\theta) = \Sigma^{-1/2}(\mathbf{y}(\theta) - \mathbf{d}).$$

Then

$$\mathcal{L}(\theta) = \frac{1}{2} \left\| \widetilde{\mathbf{r}}(\theta) \right\|^2 .$$

Likelihood turns into Euclidean geometry after whitening.

Scaled sensitivity

The whitened Jacobian is

$$\widetilde{J} = \Sigma^{-1/2} \frac{\partial \mathbf{y}}{\partial \theta}.$$

The Fisher information is

$$\mathcal{I} = \widetilde{J}^{\mathsf{T}} \widetilde{J}.$$

Thus experimental uncertainty is not an afterthought. It changes the metric on parameter space and therefore changes which directions are stiff or sloppy.

16.5 Identifiable combinations by SVD

Linearized model

Linearize a nonlinear prediction around θ_0:

$$\delta \mathbf{y} = J \delta \theta.$$

Take the SVD

$$J = \mathbf{U} \Sigma \mathbf{V}^{\mathsf{T}}.$$

Then

$$\delta \mathbf{y} = \sum_j \sigma_j \mathbf{u}_j (\mathbf{v}_j^{\mathsf{T}} \delta \theta).$$

The scalar

$$\phi_j = \mathbf{v}_j^{\mathsf{T}} \theta$$

is a locally identifiable combination if σ_j is large.

A practical interpretation

If a paper reports huge uncertainty in individual rate constants but stable predictions, do not immediately call the model bad. Ask whether the predictions depend on a few combinations. The raw parameters may be poorly identified while the scientifically relevant combinations are sharply constrained.

The second pass puts the same linear ideas inside time. An ODE is not a different subject here; it is the machine that produces $\mathbf{y}(\theta)$, and its sensitivities are the Jacobian columns we have already learned to read.

16.6 Exponential decay

One parameter

Consider

$$y(t; k) = Ae^{-kt}.$$

The sensitivity to k is

$$\frac{\partial y}{\partial k} = -tAe^{-kt}.$$

Early time points have little sensitivity to k, because t is small. Late time points may also have little sensitivity if e^{-kt} is already near zero. The most informative time scale is around $t \sim 1/k$.

Two parameters

Now let

$$y(t; A, k) = Ae^{-kt}.$$

The Jacobian row at time t_m is

$$\begin{bmatrix} e^{-kt_m} & -t_m A e^{-kt_m} \end{bmatrix}.$$

If all observations are very early, then

$$e^{-kt} \approx 1 - kt, \qquad Ae^{-kt} \approx A - Akt.$$

The data mainly see A and Ak, not A and k independently. This is a baby version of identifiable combinations.

16.7 Sums of exponentials

Why they become sloppy

For

$$y(t) = \sum_{j=1}^{N} A_j e^{-\lambda_j t},$$

nearby rates can compensate:

$$A_1 e^{-\lambda_1 t} + A_2 e^{-\lambda_2 t} \approx (A_1 + A_2) e^{-\bar{\lambda} t}$$

over a finite time window if λ_1 and λ_2 are close. Very fast rates disappear after the early transient. Very slow rates look nearly constant. These are not numerical coincidences; they are model manifold boundaries.

Connection to LTI

Every stable finite-dimensional LTI system has impulse responses made from exponentials, possibly multiplied by polynomials for repeated poles. Therefore the exponential-sum example is not a toy disconnected from systems theory. It is the scalar shadow of modal decomposition.

16.8 Sensitivity ODE

Derivation

Let

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \theta), \qquad \mathbf{x}(0) = \mathbf{x}_0(\theta).$$

Set

$$\mathbf{S}(t) = \frac{\partial \mathbf{x}(t)}{\partial \theta} \in \mathbb{R}^{n \times p}.$$

Differentiate both sides with respect to θ:

$$\dot{\mathbf{S}} = \frac{\partial \mathbf{f}}{\partial \mathbf{x}} \mathbf{S} + \frac{\partial \mathbf{f}}{\partial \theta}, \qquad \mathbf{S}(0) = \frac{\partial \mathbf{x}_0}{\partial \theta}.$$

This sensitivity system is linear in $\mathbf{S}$, even when the original ODE is nonlinear.

Why finite differences can mislead

Finite differences approximate

$$\frac{\partial \mathbf{y}}{\partial \theta_i} \approx \frac{\mathbf{y}(\theta + h\mathbf{e}_i) - \mathbf{y}(\theta)}{h}.$$

If h is too large, the approximation is nonlinear. If h is too small, floating-point cancellation dominates. Sensitivity equations avoid many of these problems, which is why MBAM implementations often prefer them for ODE models.

16.9 Two-compartment dynamics

A minimal PK-like LTI system

Consider

$$\frac{\mathrm{d}}{\mathrm{d}t} \begin{bmatrix} C_1 \\ C_2 \end{bmatrix} = \begin{bmatrix} -(k_{10} + k_{12}) & k_{21} \\ k_{12} & -k_{21} \end{bmatrix} \begin{bmatrix} C_1 \\ C_2 \end{bmatrix}.$$

The eigenvalues of the system matrix produce two exponential phases. If one phase is much faster than the sampling resolution, it may be practically invisible.

Boundary interpretation

A fast exchange limit such as

$$k_{12}, k_{21} \to \infty, \qquad \frac{k_{12}}{k_{21}} \to \rho$$

can collapse two compartments into a quasi-equilibrated effective compartment. The surviving parameter is not k_{12} or k_{21} alone, but a ratio or capacity-like combination. This is exactly the kind of limit MBAM is designed to reveal.

16.10 Spring–mass–damper

Second-order equation

The classical LTI example is

$$m\ddot{x} + b\dot{x} + kx = f(t).$$

The transfer function from force to displacement is

$$H(s) = \frac{1}{ms^2 + bs + k}.$$

The poles solve

$$ms^2 + bs + k = 0.$$

They encode damping, oscillation, and stability.

Nondimensional form

Define

$$\omega_n = \sqrt{\frac{k}{m}}, \qquad \zeta = \frac{b}{2\sqrt{mk}}.$$

Then

$$\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2 x = \frac{1}{m}f(t).$$

The three physical parameters m, b, k become combinations $\omega_n, \zeta, 1/m$. Often these combinations are more interpretable than the raw parameters.

16.11 Convolution

Impulse response

If $h(t)$ is the impulse response, any input $u(t)$ produces

$$y(t) = \int_0^t h(t-\tau)u(\tau)\,\mathrm{d}\tau.$$

Convolution says that the system remembers past input through a kernel. A fast kernel has short memory; a slow kernel has long memory.

Relevance of memory

For a QSP model, deciding whether a subsystem can be reduced often means asking: does its memory kernel matter on the query time scale? If not, it may be replaced by an algebraic relation, an effective delay, or a lower-order state.

16.12 Stiffness

Multiple time scales

An ODE is stiff when stable fast modes force tiny numerical steps even though the scientific behavior of interest evolves slowly. A simple linear example is

$$\dot{x}_1 = -x_1, \qquad \dot{x}_2 = -1000x_2.$$

The second state decays rapidly, but explicit numerical methods may still be constrained by it.

Model reduction view

Stiffness and sloppiness are not the same, but they often meet. A fast stable mode may be numerically troublesome and prediction-irrelevant after a short transient. A boundary approximation can remove it if the quantities of interest do not ask about that transient.

The third pass asks whether our computations are honest enough to support scientific interpretation. Conditioning, regularization, and stable factorization are not numerical footnotes; they decide whether a small singular value is a property of the model or an artifact of the calculation.

16.13 Finite precision

Roundoff as perturbation

Floating-point arithmetic replaces exact operations by perturbed operations. A computed solution often solves a nearby problem:

$$(\mathbf{A} + \delta\mathbf{A})\widehat{\mathbf{x}} = \mathbf{b} + \delta\mathbf{b}.$$

Backward error analysis asks whether the nearby problem is close enough to the original one.

Scientific analogy

Model reduction has a similar flavor. A reduced model is acceptable if it solves a nearby prediction problem at the precision required by the scientific question. The question is not whether the reduced model is literally identical. It is whether the discrepancy is smaller than the relevant tolerance.

16.14 Regularization

Ridge

Ridge regression solves

$$\widehat{\mathbf{x}} = \operatorname{argmin}_{\mathbf{x}} \|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2 + \lambda \|\mathbf{x}\|^2 .$$

The solution is

$$\widehat{\mathbf{x}} = (\mathbf{A}^\mathsf{T}\mathbf{A} + \lambda I)^{-1}\mathbf{A}^\mathsf{T}\mathbf{b}.$$

It suppresses directions with small singular values.

Relation to Bayesian priors

The penalty $\lambda \|\mathbf{x}\|^2$ is equivalent to a Gaussian prior on $\mathbf{x}$. In mechanistic modeling, regularization can stabilize fitting, but it does not by itself explain which mechanisms are irrelevant. MBAM aims for an explanatory reduction rather than only a stabilized inverse problem.

16.15 QR versus normal equations

Squaring the condition number

Solving least squares through

$$\mathbf{A}^{\mathsf{T}}\mathbf{A}\mathbf{x} = \mathbf{A}^{\mathsf{T}}\mathbf{b}$$

can be numerically dangerous because

$$\kappa(\mathbf{A}^{\mathsf{T}}\mathbf{A}) = \kappa(\mathbf{A})^2.$$

QR and SVD avoid this squaring.

Model-fitting moral

If a fitting problem is already ill-conditioned, careless numerical formulation can make it worse. Before interpreting parameter uncertainty scientifically, we need to know whether the uncertainty comes from the model/data geometry or from unstable computation.

The fourth pass changes the distance measure. Least squares measured Euclidean residuals; inference measures distinguishability between probability laws. Fisher information is the bridge that lets the same Jacobian language survive this change.

16.16 KL local expansion

One-dimensional sketch

Let $p(x \mid \theta)$ be smooth. The score is

$$s_\theta(x) = \frac{\partial}{\partial\theta} \log p(x \mid \theta).$$

Under regularity conditions,

$$\mathbb{E}_\theta[s_\theta(\mathsf{x})] = 0.$$

The Fisher information is

$$I(\theta) = \mathbb{E}_\theta[s_\theta(\mathsf{x})^2].$$

Then for small δ,

$$\mathrm{D}(p_\theta \| p_{\theta+\delta}) = \frac{1}{2} I(\theta)\delta^2 + o(\delta^2).$$

Why the first-order term vanishes

The first-order term vanishes because probabilities must still integrate to one:

$$\int p(x \mid \theta)\, \mathrm{d}x = 1 \quad \Rightarrow \quad \int \frac{\partial p(x \mid \theta)}{\partial\theta}\, \mathrm{d}x = 0.$$

This is the probabilistic reason Fisher information is second-order geometry.

16.17 Information projection

Projection between distributions

Information projection asks for the member of a family closest in KL divergence:

$$Q^{\star} = \operatorname{argmin}_{Q \in \mathcal{F}} \mathrm{D}(P \| Q).$$

This is not Euclidean projection, but it plays a similar role in statistical geometry.

Model reduction reading

When a reduced model family is used to approximate a full model, we can think of the reduced family as a lower-dimensional surface. The best reduced model is the projection of the full model's behavior onto that surface, according to the distinguishability relevant to the query.

16.18 Model manifold for line fitting

A simple surface

Let

$$y_i(a, b) = a + bt_i$$

for fixed time points t_i. The prediction vector is

$$\mathbf{y}(a, b) = a\mathbf{1} + b\mathbf{t}.$$

The model manifold is a plane in $\mathbb{R}^M$. The Jacobian is constant:

$$J = \begin{bmatrix} 1 & t_1 \\ \vdots & \vdots \\ 1 & t_M \end{bmatrix}.$$

Why nonlinear models are different

For nonlinear models, the tangent plane changes with θ. Small eigenvectors of the Fisher information may rotate as we move. This is why MBAM follows a geodesic instead of simply deleting the smallest local singular vector at the starting point.

The final pass stops being merely local. A sloppy direction at one point becomes useful only when we follow it until the equations reveal a boundary, a limit, or an approximation we can name.

16.19 Geodesic boundaries

Local direction, nonlocal limit

The least sensitive eigenvector is local:

$$\mathcal{I}(\theta_0)\mathbf{v}_{\min} = \lambda_{\min}\mathbf{v}_{\min}.$$

Following the geodesic reveals how that direction evolves. Near a boundary, the direction often simplifies and points toward a recognizable parameter limit.

The interpretive step

The numerical geodesic might say:

$$k_1 \to \infty, \qquad k_2 \to \infty, \qquad \frac{k_2}{k_1} \to K.$$

The modeler must then inspect the equations and take the limit carefully. The resulting reduced model is the scientific product.

16.20 Boundary taxonomy

Common limits

MBAM often finds limits such as:

$$k \to 0, \quad k \to \infty, \quad \frac{k_1}{k_2} \to K, \quad x_{\text{fast}} \to x_{\text{eq}}, \quad N \to \infty, \quad \Delta x \to 0.$$

These correspond to irreversibility, fast equilibration, quasi-steady state, thermodynamic limit, and continuum limit.

Why limits preserve meaning

Deleting a variable by numerical black-box projection may preserve output but hide mechanism. Taking a limit inside the original equations shows what physical or biological assumption is being made.

16.21 MBAM and Michaelis–Menten carefully

Mass action

For

$$E + S \underset{k_r}{\overset{k_f}{\rightleftharpoons}} C \xrightarrow{k_c} E + P,$$

mass action gives

$$\dot{C} = k_f ES - (k_r + k_c)C.$$

If binding/unbinding is fast relative to catalysis, then $k_f, k_r \to \infty$ with $K_d = k_r/k_f$ fixed. The fast equation imposes

$$k_f ES - k_r C \approx 0 \quad \Rightarrow \quad C \approx \frac{ES}{K_d}.$$

Conservation

With $E_0 = E + C$,

$$C = \frac{(E_0 - C)S}{K_d} \quad \Rightarrow \quad C(K_d + S) = E_0 S.$$

Thus

$$C = \frac{E_0 S}{K_d + S}, \qquad \dot{P} = k_c C = \frac{k_c E_0 S}{K_d + S}.$$

The reduced model keeps the combination $K_d = k_r/k_f$, not the two fast rates separately.
The last two examples close the loop back to QSP. Once the reduced mechanism exists, we still have to ask which prediction it was reduced for, and whether the decision query really ignores what we removed.

16.22 Active manifold

Training data versus query

Let $\mathcal{D}$ be calibration data and $\mathcal{Q}$ future predictions. A parameter direction can be irrelevant to $\mathcal{Q}$ even if it affects unobserved internal variables. Conversely, a direction can fit $\mathcal{D}$ weakly but matter greatly for a future intervention.

This is why reduction should be query-specific:

$$\mathcal{I}_{\mathcal{Q}} = J_{\mathcal{Q}}^{\mathsf{T}} J_{\mathcal{Q}}.$$

The active subspace

The stiff eigendirections of $\mathcal{I}_{\mathcal{Q}}$ span an active local subspace. MBAM asks whether the inactive directions can be pushed to boundaries and removed while retaining the active behavior. The result is not merely lower dimension; it is a simpler mechanistic explanation for the query.

16.23 CRISP for QSP

A workflow

For a QSP project, a CRISP-like workflow reads:

(i) Build a literature-derived mechanistic model.

(ii) Calibrate enough to make the model behave plausibly.

(iii) Define the query: virtual population, target discovery, toxicology, dose selection, or biomarker prediction.

(iv) Compute sensitivities of the query.

(v) Use information geometry to identify irrelevant combinations.

(vi) Reduce by MBAM boundaries.

(vii) Refit and validate the reduced model for the query.

Regulatory style interpretation

For model-informed decisions, a reduced QSP model should answer:

What was removed?
Why is it irrelevant?
For which prediction is it irrelevant?

This is the standard we should use in future pharmacology notes.

16.24 The full conceptual chain

From matrix to manifold

The chain is:

$$\mathbf{A} \quad \rightsquigarrow \quad J \quad \rightsquigarrow \quad J^{\mathsf{T}}J \quad \rightsquigarrow \quad \mathcal{I} \quad \rightsquigarrow \quad \text{metric} \quad \rightsquigarrow \quad \text{model manifold.}$$

Each arrow changes vocabulary but preserves a central idea: how does a small movement in one space become a movement in another space?

From ODE to MBAM

The dynamic chain is:

$$\begin{array}{c}
\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \theta) \quad \rightsquigarrow \quad \mathbf{y}(\theta) \quad \rightsquigarrow \quad \frac{\partial \mathbf{y}}{\partial \theta} \\
\downarrow \\
\text{least visible parameter direction} \quad \rightsquigarrow \quad \text{boundary limit}
\end{array}$$

That is the mathematical skeleton beneath many successful approximations in biology and pharmacology.

17 A self-study route

17.1 The order I would read

I would not start with MBAM directly. I would read in this order:

(i) 6.S087 probability and matrix tricks: random variables, covariance, gradients, Jacobians.

(ii) Matrix Cookbook: identities as a reference, not as a book to memorize.

(iii) 18.03: ODE modeling, units, linearization, qualitative behavior.

(iv) 18.031: LTI systems, convolution, transfer functions, poles.

(v) 18.335: conditioning, residual/error distinction, QR/SVD, iterative methods.

(vi) 6.441: divergence, local Fisher geometry, information projection.

(vii) MBAM papers: model manifold, geodesics, boundary limits.

(viii) QSP CRISP paper: how these ideas become a workflow for pharmacological models.

17.2 A reading checklist for any model

When reading a QSP or systems biology model, ask:

(i) What are the states $\mathbf{x}$?

(ii) What are the parameters θ?

(iii) What are the quantities of interest $\mathcal{Q}$?

(iv) What is the prediction map $\theta \mapsto \mathbf{y}(\theta)$?

(v) How is $J = \partial \mathbf{y}/\partial \theta$ computed?

(vi) What are the large and small singular directions?

(vii) Which small directions correspond to interpretable limits?

(viii) Does the reduced model preserve the prediction task, or only the training data?

(ix) What mechanistic combinations survive as control knobs?

18 Connection back to pharmacology

18.1 Why this belongs next to PK/PD

Population PK taught us that individual parameters are latent and inferred through a hierarchical model. MBAM teaches a complementary lesson: mechanistic parameters may be too detailed for the prediction task, and the meaningful objects may be combinations living on a lower-dimensional manifold.

In PK/PD terms:

$$\begin{array}{c}\text{rich mechanism}\\ \downarrow\\ \text{prediction map}\\ \downarrow\\ \text{active combinations}\\ \downarrow\\ \text{decision-ready reduced model}\end{array}$$

This is exactly what we want when moving from detailed QSP to dosing, target discovery, biomarker selection, or trial design.

18.2 The one-sentence memory hook

> MBAM is numerical linear algebra plus information geometry applied to mechanistic dynamical models: it follows the least identifiable direction until the equations reveal a scientifically meaningful limiting approximation.

18.3 Working conclusion

Linear algebra teaches how maps stretch directions. ODE teaches how biological mechanisms generate maps over time. LTI theory teaches modes, poles, convolution, and time scales. Numerical analysis teaches which computations and inverse problems are trustworthy. Information geometry teaches that distinguishability is a metric. MBAM combines these lessons into a practical question:

> Which parts of this mechanistic model are truly needed for this prediction?

That question is small enough to compute and large enough to guide QSP.